CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (IT/CE)
November 2012
IT–310 Artificial Intelligence (Elective-I)

Date: 09.11.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 **Answer the following questions.**

a) What is the role of knowledge in solving AI problems? Up to what extent and level of knowledge is required for it? **3**

b) Compare Procedural Vs Declarative knowledge. **4**

c) What is the Turing test? What is the significance of it? **4**

Q-2

[A] What do you mean by local maximum? Does Hill-Climbing technique suffer from local maximum? Give solution, if any, for solving the problem. **4**

[B] What are the characteristics of the problem that are to be analyzed when choosing an appropriate method to solve the problem? Explain. **4**

[C] Analyze 8-puzzle problem with respect to the seven problem characteristics. **4**

OR

Q-2

[A] Explain the horn clause. How is it related with PROLOG? **4**

[B] What are the conditions to be satisfied to find an optimal path to a goal, if any path to a goal exists? **4**

[C] Define an appropriate state space representation for the game of Chess. **4**

Q-3

[A] What do you mean by control strategy? What are the requirements of it? **4**

[B] Write a PROLOG Program to divide the list into odd positioned no. and even positioned no. **4**

[C] What do you mean by underestimating and overestimating a heuristic function? Under what condition(s) A^* algorithm gives optimal solution. **4**

OR

Q-3

[A] Write a PROLOG program for finding the factorial and fibonacci series. **4**

[B] Give an example of problems for which BFS would work better than DFS and vice versa. **4**

[C] Consider the following set of sentences: **4**

1. Marcus was a man
2. Marcus was a Pompeian

3. Marcus was born in 40 A.D.
4. All men are mortal
5. All pompeians died when the volcano erupted in 79 A.D.
6. No mortal lives longer than 150 years.
7. It is now 2005

Answer the following question with the help of proof by resolution:
Is marcus alive?

SECTION-II

Q-4

a) Describe in brief various steps involved in NLP (Natural Language Processing). 3

b) Give the difference between red cut and green cut with an example. 4

c) Justify the use of Bay's theorem in Bayesian N/W. 4

Q-5

[A] Draw the semantic network for the following sentence: 4
'Every dog in the town has bitten the constable.'

[B] Explain the working of Perceptron Network. 4

[C] Explain the principles of means_end_analysis approach to problem solving. 4

OR

Q-5

[A] Which factors decide the choice of reasoning forward or backward? 4

[B] How to decide about the training time required for training the artificial neural networks? 4

[C] Large Expert Systems are generally built by first implementing a prototype. Give reasons for doing this. 4

Q-6

[A] Where the knowledge in Neural Network is stored? Explain multilayer fully connected feed forward networks. 4

[B] Write a note on: Dempster Shafer Theory 4

OR

[B] Trace the execution of the constraint satisfaction procedure in solving the crypt arithmetic problem: 4

SEND
+
MORE
MONEY

[C] Compare Expert System with Neural Network and Genetic Algorithms. 4

OR

[C] Derive the parse tree for the following sentence using and writing the appropriate rules: 4
"Bill printed the .init file".